ORTHOPEDIC HOSPITAL DESIGN

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DESIGN BRIEF

- ✓Sustainability
- ✓Therapeutic
- ✓Inclusive
- ✓Accessible
- ✓Resilient

Designed in accordance with: Eurocodes and HTM's

The site is near Cardiff City Centre, bounded by buildings on 3 sides and a railway on the fourth.

This created several constraints that directly influenced the design:

- Noise from the railway
- Limited footprint
- Poor ground conditions

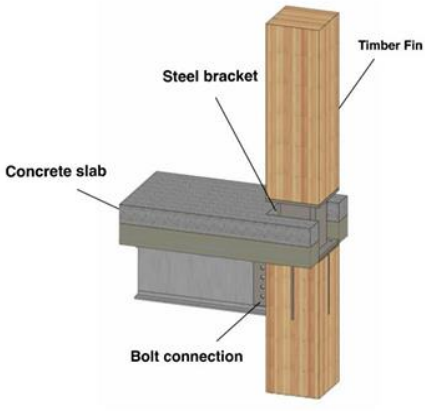
In response to this, the building was:

- Oriented to maximise daylight in patient areas
- Planned to maintain 25% green space
- Designed with separate circulation for inpatients, outpatients, staff, and services
- Served by a one-way road system to reduce congestion and improve emergency access for emergency vehicles

ARCHITECTURAL DECISIONS

The hospital is organised into three main blocks. This modular approach improves:

- Wayfinding
- Phasing potential
- Structural clarity
- Fire compartmentation



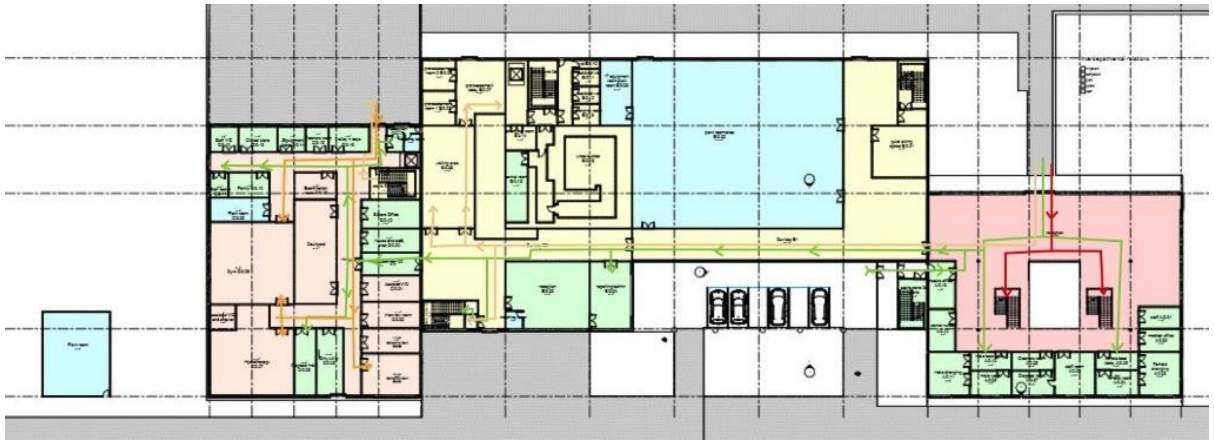
Figure–2–Façade connection

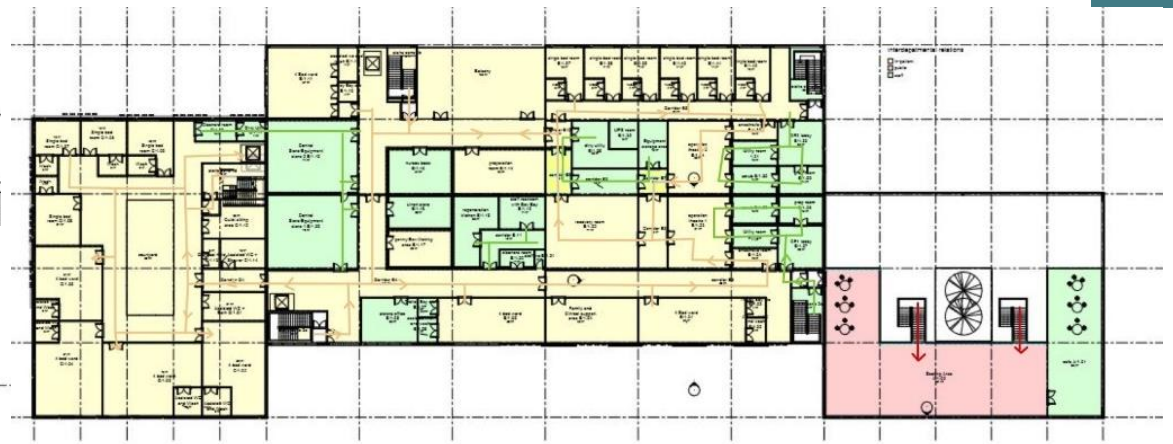
Façade Design

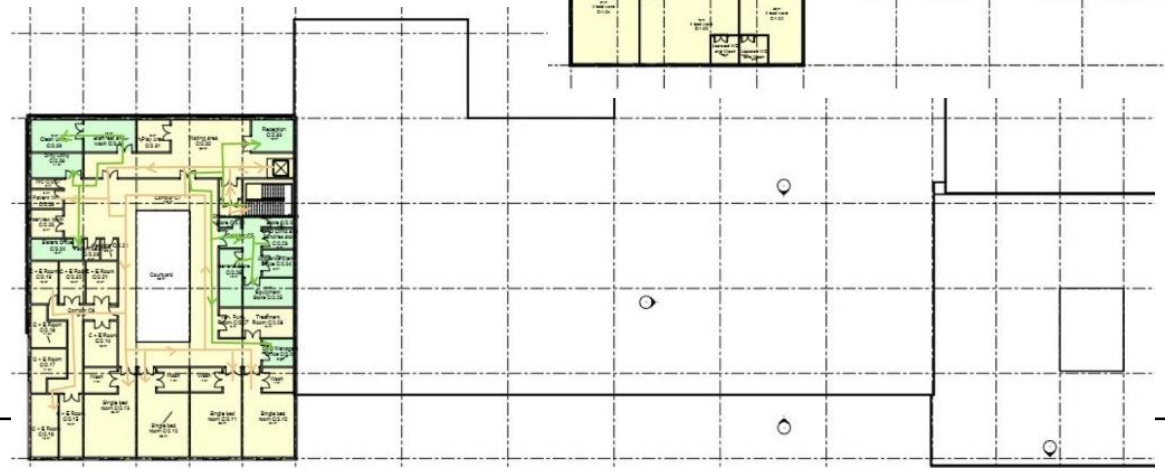
- Green walls + timber fins absorb noise and add biophilic appeal
- Cement-lime facade

ARCHITECTURAL DECISIONS

Interdepartmental Relationships:







= in-patient zones

= out-patient zones

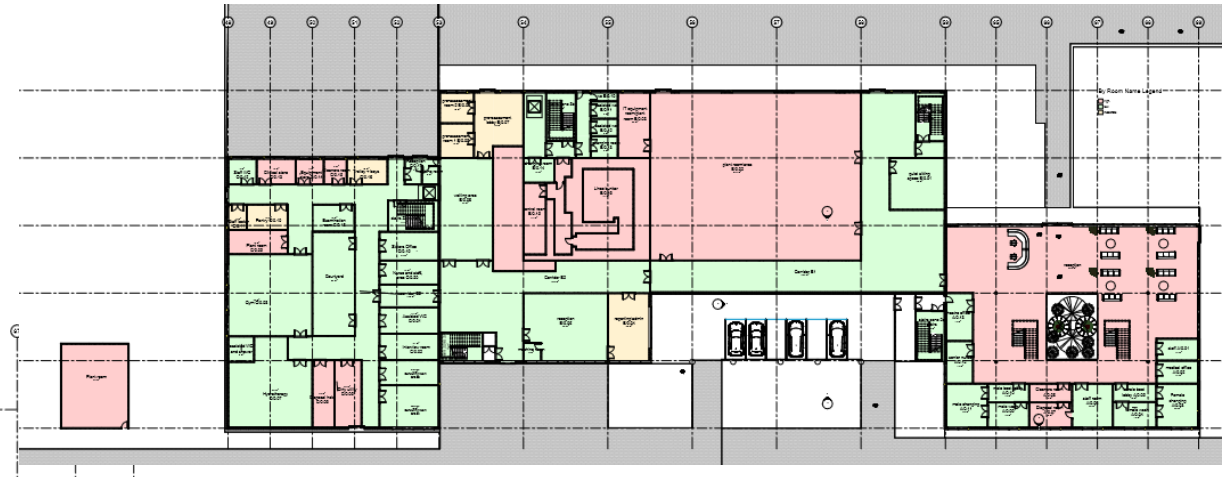
= staff only zones

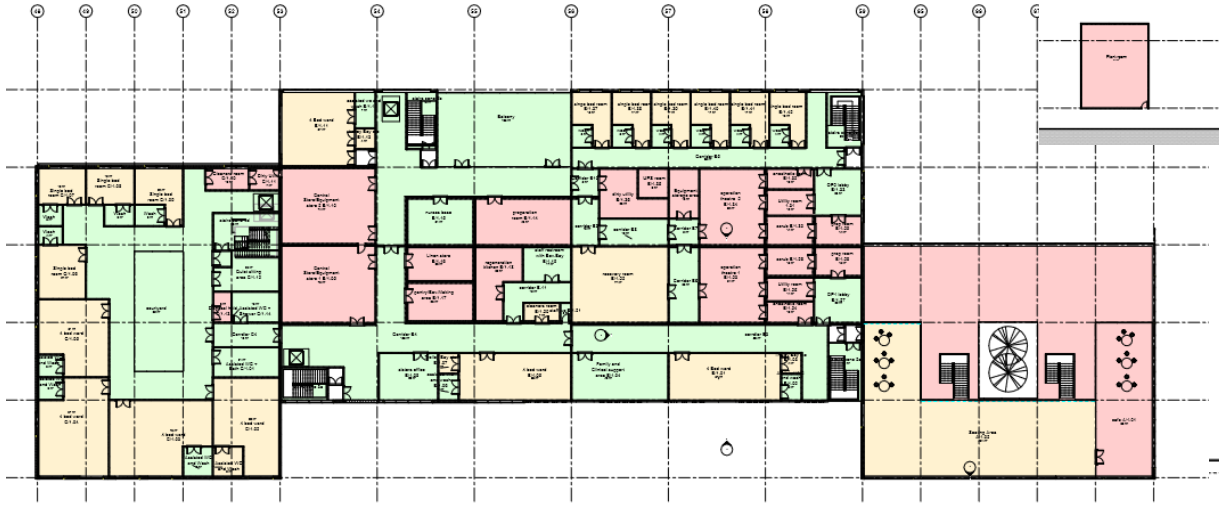
= plant rooms

= public access zones

ARCHITECTURAL DECISIONS

Fire Hazard Classification:

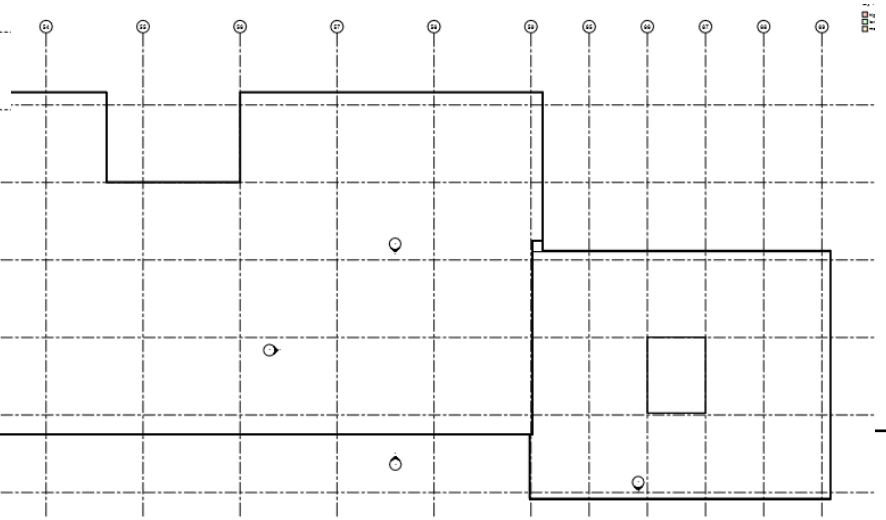




= high hazard areas

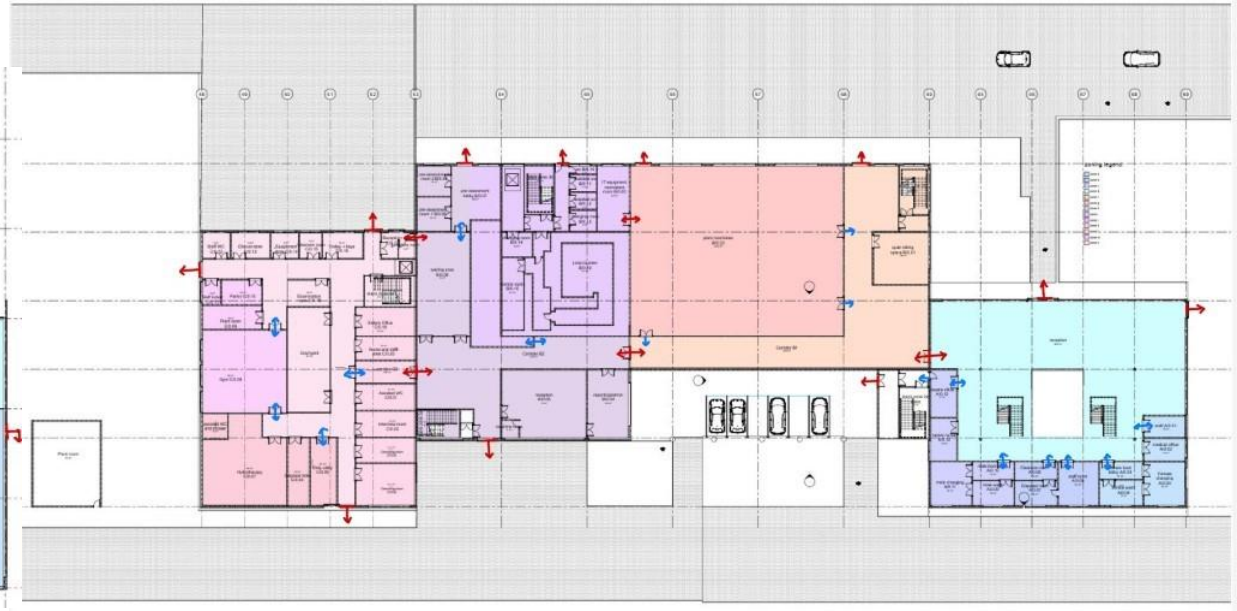
= ordinary hazard areas

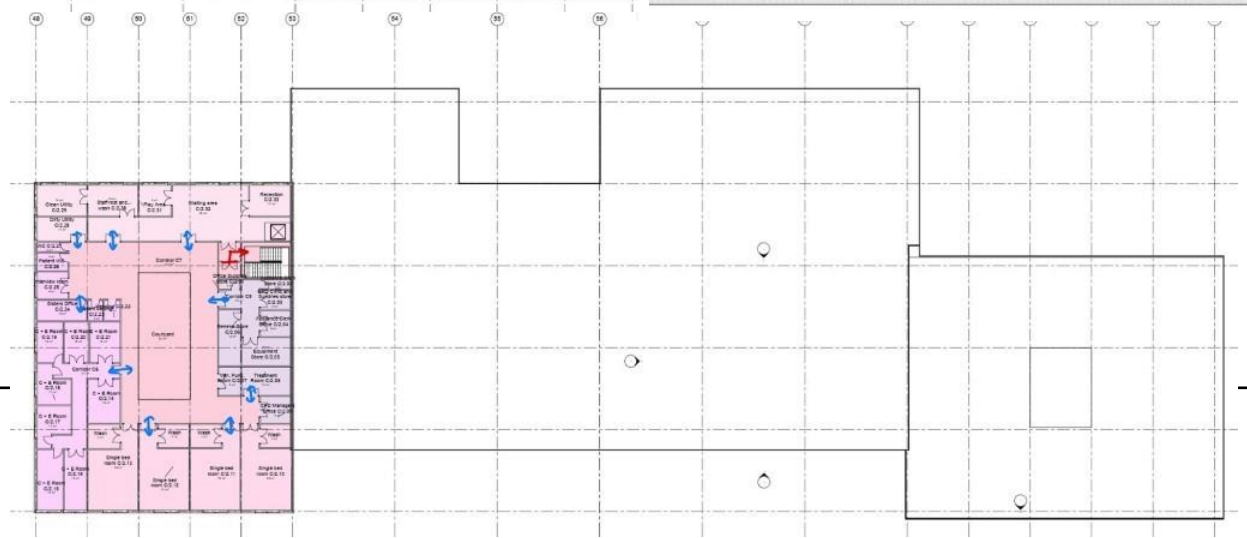
= low hazard areas



ARCHITECTURAL DECISIONS

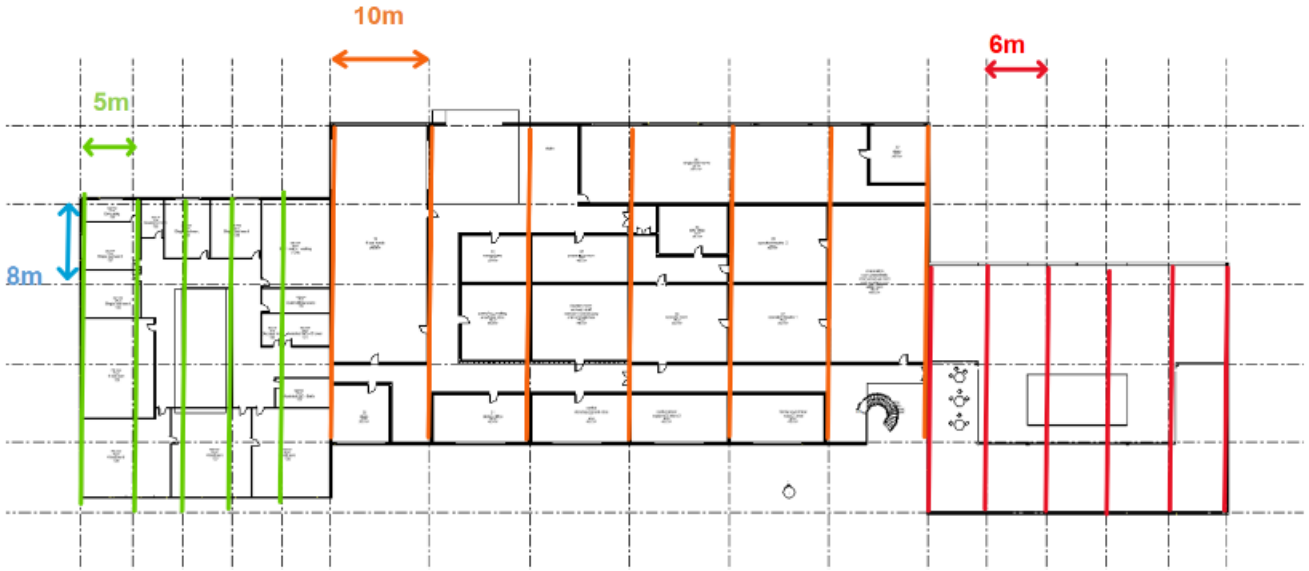
Zoning: Compartmentation and sub-compartmentation to limit fire and smoke spread - dividing the building into compartments





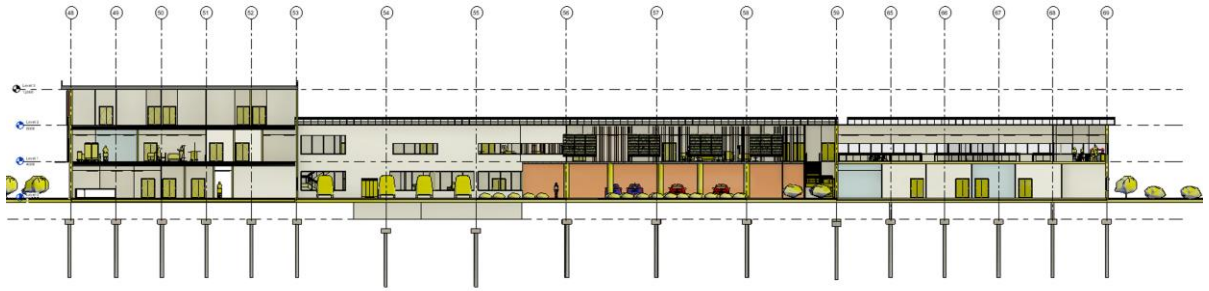
STRUCTURAL SCHEME

- Reinforced concrete structure (c/40)
- Comprised of 3 grids
  - 8m x 5m
  - 8m x 10m
  - 8m x 6m



Chosen for

- Fire resistance
- Vibration dampening

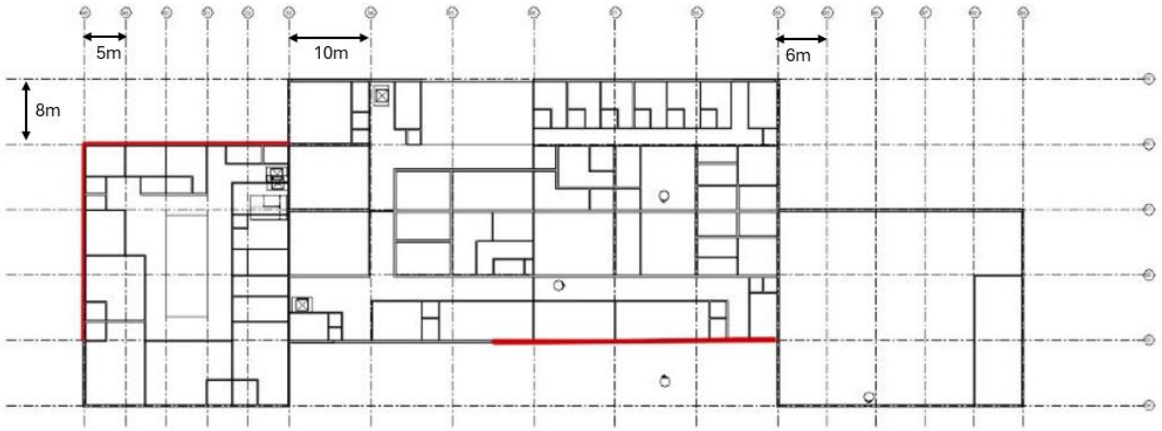


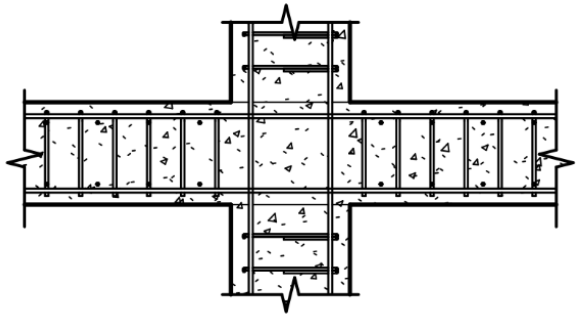
STRUCTURAL –FLAT SLABS

Flat slab size (m)	Floor	Thickness (mm)
10 x 8	First	300
10 x 8	Roof	225
8 x 6	First	200
8 x 6	Roof	175
8 x 5	First	250
8 x 5	Second	250
8 x 5	Roof	150

Flat Slabs:

- Designed based on location and floor
- Chosen for the balance of sustainability and constructability
- Punching shear as the main constraint
- Supports the façade loading

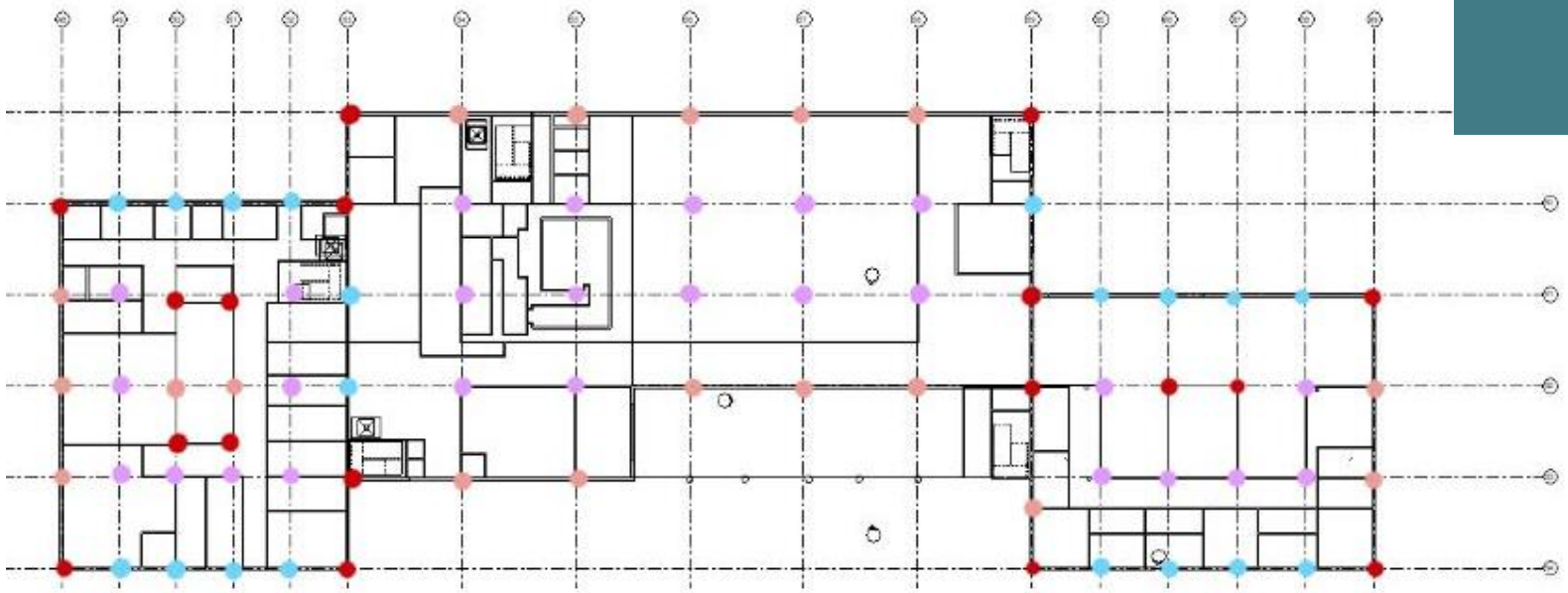




STRUCTURAL –COLUMNS

Columns:

- Used to aid punching shear
- Rectangular 200x300 mm
- Designed for based on location and floor

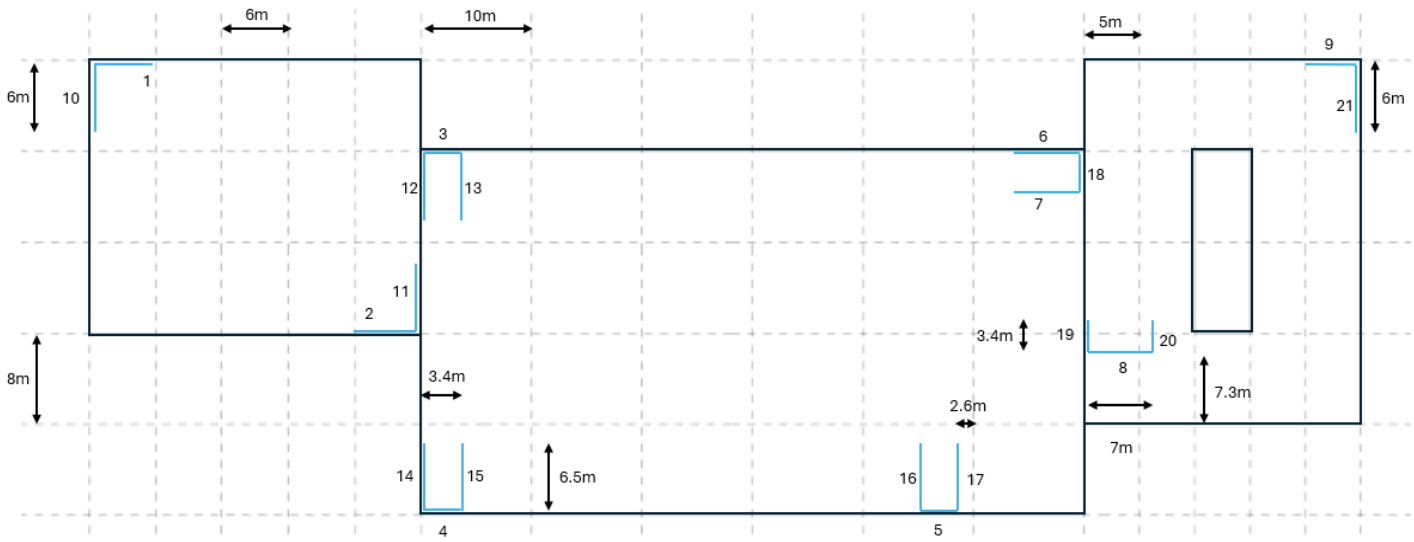


- Red – Supports slab with 2 edges
- Blue – Supports slab with 1 short edge
- Orange – supports slab with 1 long edge
- Pink – Supports internal slabs

STRUCTURAL –SHEAR

Shear wall and cores:

- Provides adequate bracing to the structure
- Surrounds stairs for fire requirements
- Designed for the local wind speeds for the site
- Sized to balance of steel reinforcement and thickness



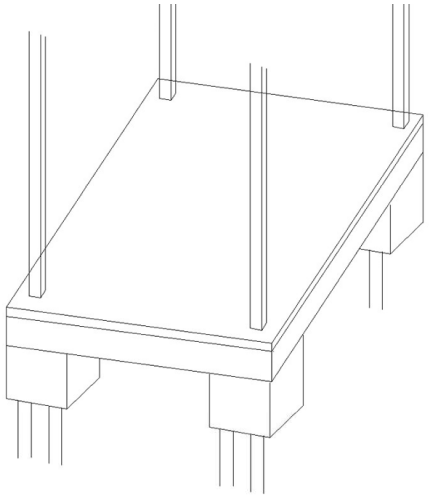
STRUCTURAL – SUSPENDED SLAB/ GROUND BEAMS

Suspended slabs

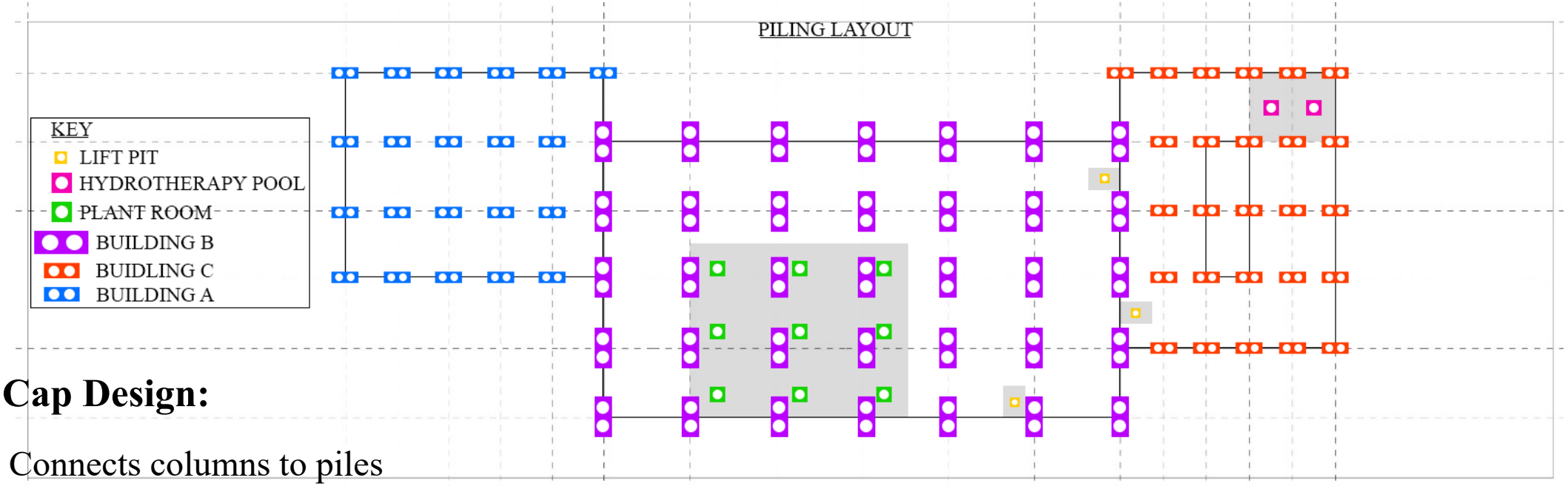
- Used across entire buildings ground floor
- Evenly distribute loads
- Minimum direct contact with weak soil

Ground beams

- Span across weak soil
- Maintain alignment
- Resist differential settlement



STRUCTURAL - FOUNDATIONS



- **Pile Cap Design:**
  - Connects columns to piles
  - Dual pile caps for primary grids
  - Single pile caps for isolated elements
  - Strut-and-Tie method for design

- **Pile Design:**
  - Continuous Flight Auger (CFA) piles
  - Minimised noise and vibration
  - Reaches Mercia Mudstone layer

THERAPEUTIC ENVIRONMENT

- The design maximises natural light and airflow, which enhances patient mood and recovery.
- Some considerations required for natural light and ventilation include windows, courtyards, daylight factors, and glare.
- A clear and simple-to-use navigation system has been used
- Inpatient and outpatient departments are separated in different buildings for easier orientation
- Courtyards have been included to boost mood and help patients recover more quickly
- Energy-efficient systems like green roofs, micro-turbines, and PV panels have created healthier spaces for patients and staff.
- Materials that emit the least carbon while maintaining their structural integrity have been chosen to extend the hospital's longevity.

Figure 1. Image of courtyards in the building

Therapy neutral Color Palette

SUSTAINABILITY

Energy modelling

- Green roof with integrated renewable energy systems like solar panels and micro-turbines.
- Maximum natural lighting by implementing windows and courtyards

Benefits of the Green Roof**:**

✓ Improved patient recovery
✓ Reduced energy costs
✓ Enhanced air quality
✓ Better stormwater management
✓ Noise reduction
✓ Extended roof lifespan.

A greywater collection system:

- Wastewater from sinks, showers, and laundry facilities is collected and reused.
- The collected greywater is repurposed for sprinklers, toilet flushing, and cleaning.

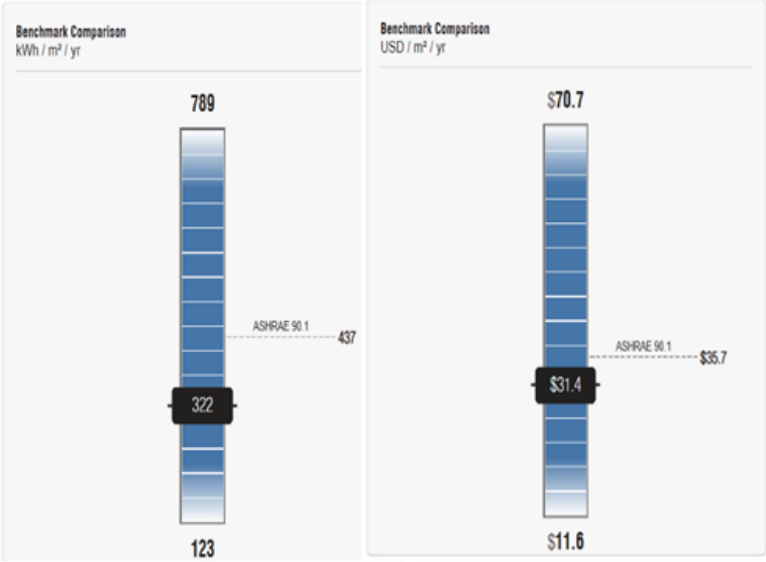
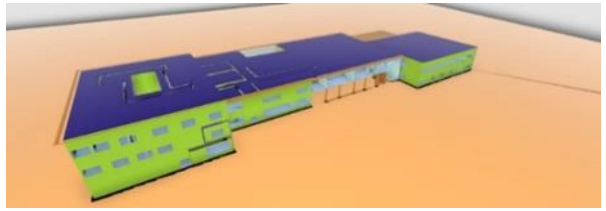


Figure 1. Energy consumption of the building and cost /m2/yr

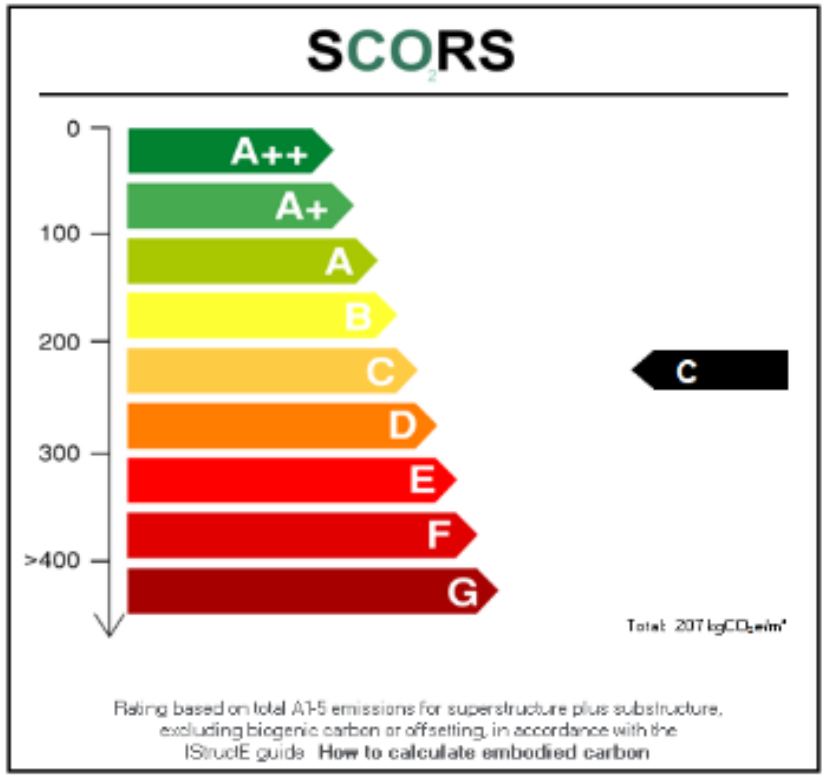


SUSTAINABILITY

Environmental Impact

- A low-carbon concrete mix reduces the structure's carbon impact.
- Recycled aggregates are incorporated into the concrete mix where possible.
- The design prioritises longevity and adaptability to extend the building's lifespan.
- Clear navigation design enhances patient experience and staff efficiency, potentially reducing energy waste from unnecessary movement

This project scheme has a SCORS rating of C



REVIT MODEL